

ADV: Financial Maths (Adv), M1 Financial Modelling (Adv)
Geometric Series (Y12)

Teacher: Ljiljana Todic

Exam Equivalent Time: 148.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

HISTORICAL CONTRIBUTION

- *M1 Financial Modelling* is a Year 12 topic that contains the substantial *Series and Application* topic of the old Mathematics course, and adds common Standard 2 content on loan and annuity tables.
- The topic is split into four sub-topics for the purposes of analysis: *1-Modelling Investments and Loans*, *2-Arithmetic Series*, *3-Geometric Series* and *4-Financial Applications of Series*.
- This analysis looks at *Geometric Series*.

HSC ANALYSIS - What to expect and common pitfalls

- *Geometric Series* must now be viewed through the prism of Financial Mathematics with the addition of the recursive definition to its content.
- *Geometric Series* accounted for an impressive 3.4% per paper in the old course (last examined in 2019). We expect this to diminish going forward due to the extra content competition in Financial Maths (notably the addition of *Modelling Investments and Loans*).
- Numerous examples using the recursive definition of a geometric sequence are included to cover syllabus changes.
- Although past HSC questions rarely involve financial themes, we have included all past HSC questions on this sub-topic in the database. These questions may involve pure calculations or non-financial applications, both of which still represent high quality revision in our view.
- Analysis of past HSC questions: Although generally well answered, note that sub-50% mean marks have resulted in 4 of the past 8 years. The summing of a series that incorporated both arithmetic and geometric elements in *2018 Adv 14d* caused major problems and warrants attention.
- Testing "sum to infinity" within the *Geometric Series* has been an examiner favourite, appearing in 7 out of the last 10 papers. The 2017 exam required a deep understanding of this concept, caused major difficulties and should be reviewed closely.

Questions

1. Financial Maths, 2ADV M1 SM-Bank 1 MC

The first three terms of a geometric sequence are **6, x , 54**.

A possible value of **x** is

- (A) **9**
 - (B) **15**
 - (C) **18**
 - (D) **24**
-

2. Financial Maths, 2ADV M1 SM-Bank 3 MC

The first three terms of a geometric sequence are

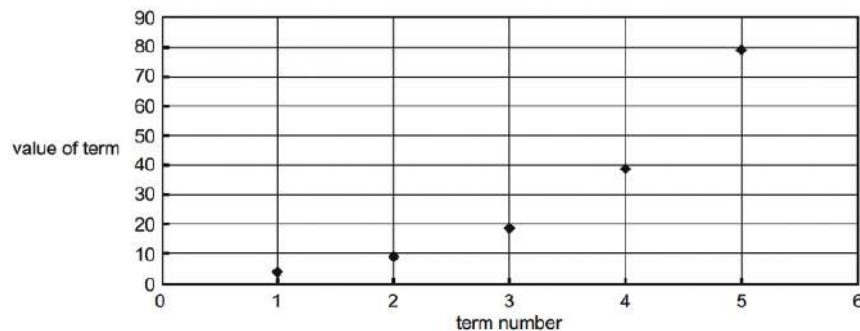
0.125, 0.25, 0.5

The fourth term in this sequence would be

- (A) **0.625**
 - (B) **0.75**
 - (C) **0.875**
 - (D) **1**
-

3. Financial Maths, 2ADV M1 SM-Bank 2 MC

The values of the first five terms of a sequence are plotted on the graph shown below.



The recursion equation that could describe the sequence is

- A. $t_{n+1} = t_n + 5$, $t_1 = 4$
- B. $t_{n+1} = 2t_n + 1$, $t_1 = 4$
- C. $t_{n+1} = t_n - 3$, $t_1 = 4$
- D. $t_{n+1} = 3t_n$, $t_1 = 4$

4. Financial Maths, 2ADV M1 SM-Bank 3 MC

The first four terms of a sequence are

12, 18, 30, 54

A recursive equation that generates this sequence is

- A. $t_{n+1} = t_n + 6$ $t_1 = 12$
- B. $t_{n+1} = 1.5t_n$ $t_1 = 12$
- C. $t_{n+1} = 0.5t_n + 12$ $t_1 = 12$
- D. $t_{n+1} = 2t_n - 6$ $t_1 = 12$

5. Financial Maths, 2ADV M1 SM-Bank 4 MC

Each trading day, a share trader buys and sells shares according to the rule

$$T_{n+1} = 0.6T_n + 50\,000$$

where T_n is the number of shares the trader owns at the start of the n th trading day.

From this rule, it can be concluded that each day

- A. the trader sells 60% of the shares that she owned at the start of the day and then buys another 50 000 shares.
- B. the trader sells 40% of the shares that she owned at the start of the day and then buys another 50 000 shares.
- C. the trader sells 50 000 of the shares that she owned at the start of the day.
- D. the trader sells 60% of the 50 000 shares that she owned at the start of the day.

6. Financial Maths, 2ADV M1 SM-Bank 5 MC

Shirley would like to purchase a new home. She will establish a loan for \$225 000 with interest charged at the rate of 3.6% per annum, compounding monthly.

Each month, Shirley will pay only the interest charged for that month.

Let V_n be the value of Shirley's loan, in dollars, after n months.

A recurrence relation that models the value of V_n is

- A. $V_0 = 225\,000$, $V_{n+1} = 1.003V_n$
- B. $V_0 = 225\,000$, $V_{n+1} = 1.036V_n$
- C. $V_0 = 225\,000$, $V_{n+1} = 1.003V_n - 8100$
- D. $V_0 = 225\,000$, $V_{n+1} = 1.003V_n - 675$

7. Financial Maths, 2ADV M1 2014 HSC 8 MC

Which expression is a term of the geometric series $3x - 6x^2 + 12x^3 - \dots$?

- (A) $3072x^{10}$
- (B) $-3072x^{10}$
- (C) $3072x^{11}$
- (D) $-3072x^{11}$

8. Financial Maths, 2ADV M1 SM-Bank 2 MC

The first four terms of a geometric sequence are 6400, t_2 , 8100, -9112.5

The value of t_2 is

- (A) -7250
- (B) -7200
- (C) -1700
- (D) 7200

9. Financial Maths, 2ADV M1 SM-Bank 4 MC

The first four terms of a geometric sequence are

4, -8 , 16, -32

The sum of the first ten terms of this sequence is

- (A) -2048
- (B) -1364
- (C) **684**
- (D) **1367**

10. Financial Maths, 2ADV M1 SM-Bank 7 MC

A dragster is travelling at a speed of 100 km/h.

It increases its speed by

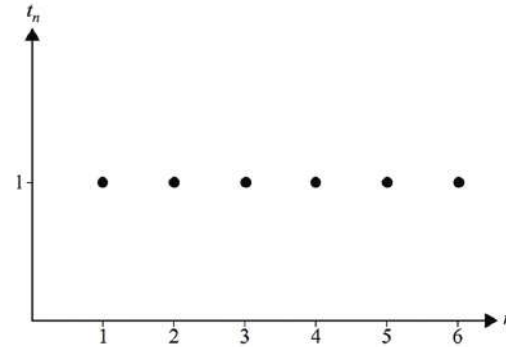
- 50 km/h in the 1st second
- 30 km/h in the 2nd second
- 18 km/h in the 3rd second

and so on in this pattern.

Correct to the nearest whole number, the greatest speed, in km/h, that the dragster will reach is

- (A) **125**
- (B) **200**
- (C) **220**
- (D) **225**

11. Financial Maths, 2ADV M1 SM-Bank 10 MC

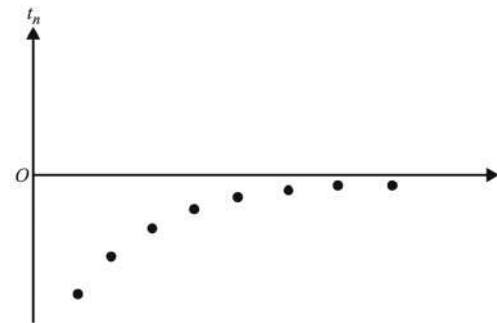


The graph above shows the first six terms of a sequence.

This sequence could be

- (A) an arithmetic sequence that sums to one.
- (B) an arithmetic sequence with a common difference of one.
- (C) a geometric sequence with an infinite sum of one.
- (D) a geometric sequence with a common ratio of one.

12. Financial Maths, 2ADV M1 SM-Bank 9 MC



The graph above shows consecutive terms of a sequence.

The sequence could be

- (A) geometric with common ratio r , where $r < 0$
- (B) geometric with common ratio r , where $0 < r < 1$
- (C) geometric with common ratio r , where $r > 1$
- (D) arithmetic with common difference d , where $d < 0$

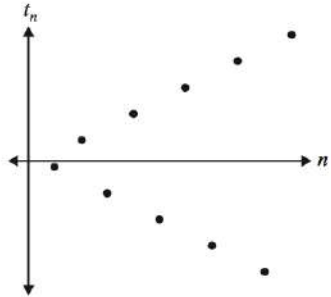
13. Financial Maths, 2ADV M1 SM-Bank 13 MC

The first term of a geometric sequence is a , where $a < 0$.

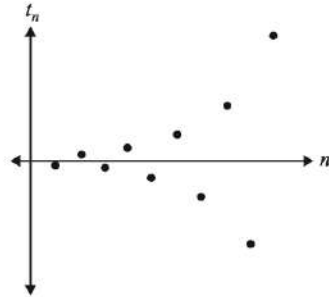
The common ratio of this sequence, r , is such that $r < -1$.

Which one of the following graphs best shows the first 10 terms of this sequence?

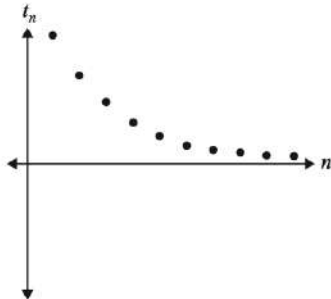
A.



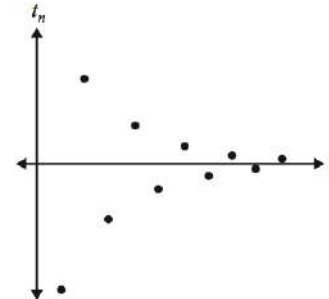
B.



C.



D.



14. Financial Maths, 2ADV M1 2009 HSC 2c

Evaluate $\sum_{k=1}^4 (-1)^k k^2$. (2 marks)

15. Financial Maths, 2ADV M1 2009 HSC 4a

A tree grows from ground level to a height of 1.2 metres in one year. In each subsequent year, it grows $\frac{9}{10}$ as much as it did in the previous year.

Find the limiting height of the tree. (2 marks)

16. Financial Maths, 2ADV M1 2010 HSC 1f

Find the limiting sum of the geometric series ..

$$1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} \dots \quad (2 \text{ marks})$$

17. Financial Maths, 2ADV M1 2006 HSC 1f

Find the limiting sum of the geometric series $\frac{13}{5} + \frac{13}{25} + \frac{13}{125} + \dots$ (2 marks)

18. Financial Maths, 2ADV M1 2006 HSC 3b

Evaluate $\sum_{r=2}^4 \frac{1}{r}$. (1 mark)

19. Financial Maths, 2ADV M1 2007 HSC 1d

Find the limiting sum of the geometric series

$$\frac{3}{4} + \frac{3}{16} + \frac{3}{64} + \dots \quad (2 \text{ marks})$$

20. Financial Maths, 2ADV M1 2015 HSC 11d

Find the limiting sum of the geometric series $1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots$ (2 marks)

21. Financial Maths, 2ADV M1 2019 HSC 11d

What is the limiting sum of the following geometric series?

$$2000 - 1200 + 720 - 432 \dots \quad (2 \text{ marks})$$

22. Financial Maths, 2ADV M1 SM-Bank 15

Phil is a builder who has purchased a large set of tools.

The value of Phil's tools is depreciated using the reducing balance method.

The value of the tools, in dollars, after n years, V_n , can be modelled by the recurrence relation shown below.

$$V_0 = 60\,000, \quad V_{n+1} = 0.9V_n$$

- Use recursion to show that the value of the tools after two years. (1 mark)
- Phil plans to replace these tools when their value first falls below \$20 000.
After how many years will Phil replace these tools? (1 mark)
- Phil has another option for depreciation. He depreciates the value of the tools by a flat rate of 8% of the purchase price per annum.
Let V_n be the value of the tools after n years, in dollars.
Write down a recurrence relation, in terms of V_0 , V_{n+1} and V_n , that could be used to model the value of the tools using this flat rate depreciation. (1 mark)

23. Financial Maths, 2ADV M1 SM-Bank 6

Julie deposits some money into a savings account that will pay compound interest every month.

The balance of Julie's account, in dollars, after n months, V_n , can be modelled by the recurrence relation shown below.

$$V_0 = 12\,000, \quad V_{n+1} = 1.0062 V_n$$

- Recursion can be used to calculate the balance of the account after one month.
 - Write down a calculation to show that the balance in the account after one month, V_1 , is \$12 074.40. (1 mark)
 - After how many months will the balance of Julie's account first exceed \$12 300 (1 mark)
- A rule of the form $V_n = a \times b^n$ can be used to determine the balance of Julie's account after n months.
 - Complete this rule for Julie's investment after n months by writing the appropriate numbers in the boxes provided below. (1 mark)

$$\text{balance} = \boxed{} \times \boxed{}^n$$

- What would be the value of n if Julie wanted to determine the value of her investment after three years? (1 mark)

24. Financial Maths, 2ADV M1 2008 HSC 4b

The zoom function in a software package multiplies the dimensions of an image by 1.2. In an image, the height of a building is 50 mm. After the zoom function is applied once, the height of the building in the image is 60 mm. After the second application, it is 72 mm.

- Calculate the height of the building in the image after the zoom function has been applied eight times. Give your answer to the nearest mm. (2 marks)
- The height of the building in the image is required to be more than 400 mm. Starting from the original image, what is the least number of times the zoom function must be applied? (2 marks)

25. Financial Maths, 2ADV M1 2012 HSC 15a

Rectangles of the same height are cut from a strip and arranged in a row. The first rectangle has width 10cm. The width of each subsequent rectangle is 96% of the width of the previous rectangle.



- Find the length of the strip required to make the first ten rectangles. (2 marks)
- Explain why a strip of 3m is sufficient to make any number of rectangles. (1 mark)

26. Financial Maths, 2ADV M1 2005 HSC 7a

Anne and Kay are employed by an accounting firm.

Anne accepts employment with an initial annual salary of \$50 000. In each of the following years her annual salary is increased by \$2500.

Kay accepts employment with an initial annual salary of \$50 000. In each of the following years her annual salary is increased by 4%.

- What is Anne's annual salary in her thirteenth year? (2 marks)
- What is Kay's annual salary in her thirteenth year? (2 marks)
- By what amount does the total amount paid to Kay in her first twenty years exceed that paid to Anne in her first twenty years? (3 marks)

27. Financial Maths, 2ADV M1 2014 HSC 14d

At the beginning of every 8-hour period, a patient is given 10 mL of a particular drug.

During each of these 8-hour periods, the patient's body partially breaks down the drug. Only $\frac{1}{3}$ of the total amount of the drug present in the patient's body at the beginning of each 8-hour period remains at the end of that period.

- How much of the drug is in the patient's body immediately after the second dose is given? (1 mark)
- Show that the total amount of the drug in the patient's body never exceeds 15 mL. (2 marks)

28. Financial Maths, 2ADV M1 SM-Bank 8

When placed in a pond, the length of a fish was 14.2 centimetres.

During its first month in the pond, the fish increased in length by 3.6 centimetres.

During its n th month in the pond, the fish increased in length by G_n centimetres, where $G_{n+1} = 0.75G_n$

Calculate the maximum length this fish can grow to. (3 marks)

29. Financial Maths, 2ADV M1 2004 HSC 7a

Evaluate $\sum_{n=2}^4 n^2$. (1 mark)

30. Financial Maths, 2ADV M1 SM-Bank 11

Express the recurring decimal $0.323232\dots$ as a fraction. (2 marks)

31. Financial Maths, 2ADV M1 2018 HSC 14d

An artist posted a song online. Each day there were $2^n + n$ downloads, where n is the number of days after the song was posted.

- Find the number of downloads on each of the first 3 days after the song was posted. (1 mark)
 - What is the total number of times the song was downloaded in the first 20 days after it was posted? (2 marks)
-

32. Financial Maths, 2ADV M1 2008 HSC 5b

Consider the geometric series

$$5 + 10x + 20x^2 + 40x^3 + \dots$$

- For what values of x does this series have a limiting sum? (2 marks)
 - The limiting sum of this series is 100.
Find the value of x . (2 marks)
-

33. Financial Maths, 2ADV M1 2011 HSC 5a

The number of members of a new social networking site doubles every day. On Day 1 there were 27 members and on Day 2 there were 54 members.

- How many members were there on Day 12? (1 mark)
 - On which day was the number of members first greater than 10 million? (2 marks)
 - The site earns 0.5 cents per member per day. How much money did the site earn in the first 12 days? Give your answer to the nearest dollar. (2 marks)
-

34. Financial Maths, 2ADV M1 2013 HSC 12c

Kim and Alex start jobs at the beginning of the same year. Kim's annual salary in the first year is \$30,000 and increases by 5% at the beginning of each subsequent year. Alex's annual salary in the first year is \$33,000, and increases by \$1,500 at the beginning of each subsequent year.

- Show that in the 10th year, Kim's annual salary is higher than Alex's annual salary. (2 marks)
 - In the first 10 years how much, in total, does Kim earn? (2 marks)
 - Every year, Alex saves $\frac{1}{3}$ of her annual salary. How many years does it take her to save \$87,500? (3 marks)
-

35. Trigonometry, 2ADV T2 2004 HSC 9a

Consider the geometric series $1 - \tan^2 \theta + \tan^4 \theta - \dots$

- When the limiting sum exists, find its value in simplest form. (2 marks)
- For what values of θ in the interval

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2} \text{ does the limiting sum of the series exist? (2 marks)}$$

36. Financial Maths, 2ADV M1 2016 HSC 14d

By summing the geometric series $1 + x + x^2 + x^3 + x^4$, or otherwise,

$$\text{find } \lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1}. \text{ (2 marks)}$$

37. Financial Maths, 2ADV M1 2017 HSC 16b

A geometric series has first term a and limiting sum 2.

Find all possible values for a . (3 marks)

38. Financial Maths, 2ADV M1 2011 HSC 9d

- Rationalise the denominator in the expression $\frac{1}{\sqrt{n} + \sqrt{n+1}}$ where n is an integer and $n \geq 1$. (1 mark)

- Using your result from part (i), or otherwise, find the value of the sum

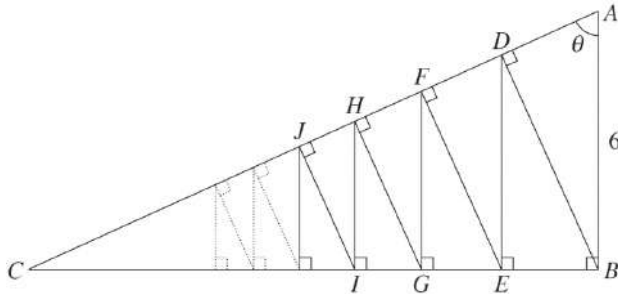
$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{99} + \sqrt{100}} \text{ (2 marks)}$$

39. Probability, 2ADV S1 2013 HSC 15d

Pat and Chandra are playing a game. They take turns throwing two dice. The game is won by the first player to throw a double six. Pat starts the game.

- Find the probability that Pat wins the game on the first throw. (1 mark)
- What is the probability that Pat wins the game on the first or on the second throw? (2 marks)
- Find the probability that Pat eventually wins the game. (2 marks)

40. Trigonometry, 2ADV T1 2005 HSC 9b



The triangle ABC has a right angle at B , $\angle BAC = \theta$ and $AB = 6$. The line BD is drawn perpendicular to AC . The line DE is then drawn perpendicular to BC . This process continues indefinitely as shown in the diagram.

- Find the length of the interval BD , and hence show that the length of the interval EF is $6 \sin^3 \theta$. (2 marks)
- Show that the limiting sum

$$BD + EF + GH + \dots$$

is given by $6 \sec \theta \tan \theta$. (3 marks)

Worked Solutions

1. Financial Maths, 2ADV M1 SM-Bank 1 MC

Since the sequence is geometric,

$$r = \frac{x}{6} = \frac{54}{x}$$

$$\begin{aligned} x^2 &= 54 \times 6 \\ &= 324 \end{aligned}$$

$$\begin{aligned} \therefore x &= 18 \\ &\Rightarrow C \end{aligned}$$

2. Financial Maths, 2ADV M1 SM-Bank 3 MC

GP sequence is 0.125, 0.25, 0.5

$$a = 0.125$$

$$r = \frac{t_2}{t_1} = \frac{0.25}{0.125} = 2$$

$$\begin{aligned} T_4 &= ar^3 \\ &= 0.125 \times 2^3 \\ &= 1 \\ &\Rightarrow D \end{aligned}$$

3. Financial Maths, 2ADV M1 SM-Bank 2 MC

By elimination,

There is no common difference between terms,

$\therefore$ Cannot be A, or C

The equation in B has $t_2 = 9$, while the equation in C has $t_2 = 12$.

$\Rightarrow B$

4. Financial Maths, 2ADV M1 SM-Bank 3 MC

Calculating t_3 in each given option

eliminates A, B and C.

$\Rightarrow D$

5. Financial Maths, 2ADV M1 SM-Bank 4 MC

$$T_{n+1} = 0.6 T_n + 50000$$

The difference equation describes a rule where a trader sells 40% of shares owned on the day before (left with 60% or $0.6 T_n$) and then buys another 50 000 each day.

$\Rightarrow B$

6. Financial Maths, 2ADV M1 SM-Bank 5 MC

Monthly interest rate

$$= \frac{3.6\%}{12} = 0.3\% = 0.003$$

Monthly payment

$$= 225\,000 \times 0.3\% \\ = \$675$$

$\therefore$ Recurrence Relation is

$$V_{n+1} = 1.003V_n - 675$$

$\Rightarrow D$

7. Financial Maths, 2ADV M1 2014 HSC 8 MC

$$3x - 6x^2 + 12x^3 - \dots$$

$$a = 3x$$

$$r = \frac{T_2}{T_1} = \frac{-6x^2}{3x} = -2x$$

$$\therefore T_n = ar^n = 3x(-2x)^n \\ = 3(-2)^n x^{n+1}$$

If $n = 9$

$$T_9 = 3(-2)^9 x^{9+1} = -1536x^{10}$$

If $n = 10$

$$T_{10} = 3(-2)^{10} x^{10+1} = 3072x^{11}$$

$\Rightarrow C$

8. Financial Maths, 2ADV M1 SM-Bank 2 MC

GP is 6400, t_2 , 8100, -9112.5

$$r = \frac{t_2}{t_1} = \frac{t_3}{t_2}$$

$$\therefore \frac{t_2}{6400} = \frac{-9112.5}{8100}$$

$$t_2 = \frac{-9112.5 \times 6400}{8100}$$

$$= -7200$$

$\Rightarrow B$

9. Financial Maths, 2ADV M1 SM-Bank 4 MC

4, -8 , 16, -32 , ...

GP where $a = 4$

$$r = \frac{t_2}{t_1} = \frac{-8}{4} = -2$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\therefore S_{10} = \frac{4[(-2)^{10} - 1]}{-2 - 1}$$

$$= -1364$$

$\Rightarrow B$

10. Financial Maths, 2ADV M1 SM-Bank 7 MC

Sequence of speed increases is

50, 30, 18, ...

GP where $a = 50$, and

$$r = \frac{t_2}{t_1} = \frac{30}{50} = 0.6$$

Since $|r| < 1$,

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{50}{1-0.6} \\ &= 125 \end{aligned}$$

$\therefore$ Max speed is $100 + 125 = 225$ km/h

$\Rightarrow D$

11. Financial Maths, 2ADV M1 SM-Bank 10 MC

Series is 1, 1, 1, ...

The only possibility within the choices

is a geometric sequence where $r = 1$.

$\Rightarrow D$

12. Financial Maths, 2ADV M1 SM-Bank 9 MC

As n increases, $t_n \rightarrow 0$.

$\therefore$ Series is a GP with a limiting sum,

i.e. $|r| < 1$.

Only one choice satisfies these conditions.

$\Rightarrow B$

13. Financial Maths, 2ADV M1 SM-Bank 13 MC

Using elimination

$a < 0$. Cannot be C.

$r < -1$. Successive terms change sign and increase exponentially.

Cannot be A, D.

$\Rightarrow B$

14. Financial Maths, 2ADV M1 2009 HSC 2c

$$\begin{aligned} \sum_{k=1}^4 (-1)^k k^2 &= (-1)^1 \times 1^2 + (-1)^2 \times 2^2 + (-1)^3 \times 3^2 + (-1)^4 \times 4^2 \\ &= -1 + 4 - 9 + 16 \\ &= 10 \end{aligned}$$

15. Financial Maths, 2ADV M1 2009 HSC 4a

$$a = 1.2, \quad r = \frac{9}{10}$$

Since $|r| < 1$,

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{1.2}{1 - \left(\frac{9}{10}\right)} \\ &= 12 \text{ m} \end{aligned}$$

$\therefore$ Limiting height of tree is 12 m.

16. Financial Maths, 2ADV M1 2010 HSC 1f

$$a = 1$$

$$r = \frac{T_2}{T_1} = \frac{-\frac{1}{3}}{1} = -\frac{1}{3}$$

Since $|r| < 1$,

$$\begin{aligned}\therefore S_{\infty} &= \frac{a}{1-r} \\ &= \frac{1}{1 - (-\frac{1}{3})} \\ &= \frac{3}{4}\end{aligned}$$

17. Financial Maths, 2ADV M1 2006 HSC 1f

$$\frac{13}{5} + \frac{13}{25} + \frac{13}{125}$$

$\Rightarrow$ GP where $a = \frac{13}{5}$, and

$$r = \frac{T_2}{T_1} = \frac{13}{25} \div \frac{13}{5} = \frac{1}{5}$$

Since $|r| < 1$

$$\begin{aligned}S_{\infty} &= \frac{a}{1-r} \\ &= \frac{\frac{13}{5}}{1 - \frac{1}{5}} \\ &= \frac{13}{5} \times \frac{5}{4} \\ &= \frac{13}{4}\end{aligned}$$

18. Financial Maths, 2ADV M1 2006 HSC 3b

$$\begin{aligned}\sum_{r=2}^4 \frac{1}{r} &= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \\ &= \frac{13}{12}\end{aligned}$$

19. Financial Maths, 2ADV M1 2007 HSC 1d

$$\frac{3}{4} + \frac{3}{16} + \frac{3}{64} + \dots$$

$\Rightarrow$ GP where $a = \frac{3}{4}$, $r = \frac{T_2}{T_1} = \frac{1}{4}$

$$\begin{aligned}\therefore S_{\infty} &= \frac{a}{1-r} \\ &= \frac{\frac{3}{4}}{1 - \frac{1}{4}} \\ &= 1\end{aligned}$$

20. Financial Maths, 2ADV M1 2015 HSC 11d

$$1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots$$

$r = -\frac{1}{4}$, $a = 1$

Since $|r| = \frac{1}{4} < 1$

$$\begin{aligned}S_{\infty} &= \frac{a}{1-r} \\ &= \frac{1}{1 - (-\frac{1}{4})} \\ &= \frac{1}{\frac{5}{4}} \\ &= \frac{4}{5}\end{aligned}$$

21. Financial Maths, 2ADV M1 2019 HSC 11d

$$\text{GP: } r = \frac{T_2}{T_1} = \frac{-1200}{2000} = -\frac{3}{5}$$

$$|r| < 1$$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{2000}{1 + \frac{3}{5}} \\ &= 1250 \end{aligned}$$

22. Financial Maths, 2ADV M1 SM-Bank 15

a. $V_0 = 60\,000$

$$V_1 = 0.9 \times 60\,000 = 54\,000$$

$$V_2 = 0.9 \times 54\,000 = \$48\,600$$

b. Find n such that:

$$60\,000 \times 0.9^n < 20\,000$$

$$\log_e 0.9^n < \log_e \left(\frac{1}{3}\right)$$

$$\begin{aligned} n &> \frac{\log_e \left(\frac{1}{3}\right)}{\log_e 0.9} \\ &> 10.427 \end{aligned}$$

$\therefore$ Phil will replace in the 11th year.

c. Annual depreciation = $0.08 \times V_0$

$$\therefore V_{n+1} = V_n - 0.08V_0$$

COMMENT: Dividing by $\log_e 0.9$
is dividing by a negative number!

23. Financial Maths, 2ADV M1 SM-Bank 6

a.i. $V_1 = 1.0062 \times V_0$

$$= 1.0062 \times 12000$$

$$= \$12\,074.40 \dots \text{ as required.}$$

a.ii. $V_2 = 1.0062 \times 12\,074.40 = 12\,149.26$

$$V_3 = 1.0062 \times 12\,149.26 = 12\,224.59$$

$$V_4 = 1.0062 \times 12\,224.59 = 12\,300.38$$

$\therefore$ After 4 months

b.i. balance = $12\,000 \times 1.0062^n$

b.ii. $n = 12 \times 3 = 36$

24. Financial Maths, 2ADV M1 2008 HSC 4b

i. $T_1 = a = 50$

$$T_2 = ar^1 = 50(1.2) = 60$$

$$T_3 = ar^2 = 50(1.2)^2 = 72$$

$$\Rightarrow \text{GP where } a = 50, r = 1.2$$

$\vdots$

$$T_9 = 50(1.2)^8$$

$$= 214.99$$

$\therefore$ Height will be 215 mm (nearest mm)

ii. $T_n = ar^{n-1} > 400$

$$\therefore 50(1.2)^{n-1} > 400$$

$$1.2^{n-1} > 8$$

$$\ln 1.2^{n-1} > \ln 8$$

$$n - 1 > \frac{\ln 8}{\ln 1.2}$$

$$n > 12.405$$

$\therefore$ The height of the building in the 13th image will be higher than 400 mm, which is the 12th time the zoom would be applied.

MARKER'S COMMENT: Within this GP, note that T_9 is the term where the zoom has been applied 8 times.

25. Financial Maths, 2ADV M1 2012 HSC 15a

i. $T_1 = a = 10$

$$T_2 = ar = 10 \times 0.96 = 9.6$$

$$T_3 = ar^2 = 10 \times 0.96^2 = 9.216$$

$$\Rightarrow \text{GP where } a = 10 \text{ and } r = 0.96$$

S_{10} = Length of strip for 10 rectangles

$$= \frac{a(1 - r^n)}{1 - r}$$

$$= 10 \left(\frac{1 - 0.96^{10}}{1 - 0.96} \right)$$

$$= 83.8 \text{ cm (to 1 d.p.)}$$

MARKER'S COMMENT: A common error was to find T_{10} instead of S_{10}

ii. Since $|r| < 1$

$$S_{\infty} = \frac{a}{1 - r}$$

$$= \frac{10}{1 - 0.96}$$

$$= 250 \text{ cm}$$

$\therefore$ Since 3 m > 2.5 m, it is sufficient.

26. Financial Maths, 2ADV M1 2005 HSC 7a

i. Let T_n = Anne's salary in year n

$$T_1 = a = \$50\,000$$

$$T_2 = a + d = \$52\,500$$

$$\Rightarrow \text{AP where } a = \$50\,000, \quad d = \$2500$$

$$T_n = a + (n - 1)d$$

$$\begin{aligned} T_{13} &= 50\,000 + (13 - 1) \times 2500 \\ &= 80\,000 \end{aligned}$$

$\therefore$ Anne's salary in her 13th year is \$80 000.

ii. Let K_1 = Kay's salary in year n

$$K_1 = a = 50\,000$$

$$K_2 = ar = 50\,000 \times 1.04 = 52\,000$$

$$\Rightarrow \text{GP where } a = 50\,000, \quad r = 1.04$$

$$K_n = ar^{n-1}$$

$$K_{13} = 50\,000 \times (1.04)^{12}$$

$$= \$80\,051.61\dots$$

$$= \$80\,052 \quad (\text{nearest dollar})$$

iii. Anne

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\begin{aligned} S_{20} &= \frac{20}{2}[2 \times 50\,000 + (20 - 1)2500] \\ &= 10[100\,000 + 47\,500] \\ &= \$1\,475\,000 \end{aligned}$$

Kay

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\begin{aligned} S_{20} &= \frac{50\,000(1.04^{20} - 1)}{1.04 - 1} \\ &= \$1\,488\,903.929\dots \end{aligned}$$

Difference

$$= 1\,488\,903.929\dots - 1\,475\,000$$

$$= \$13\,903.928\dots$$

$$= \$13\,904 \quad (\text{nearest \$})$$

$\therefore$ Kay's total salary exceeds Anne's by \$13 904

27. Financial Maths, 2ADV M1 2014 HSC 14d

i. Let A = Amount of drug in body

Initially $A = 10$

$$\text{After 8 hours } A = \frac{1}{3} \times 10$$

$$\begin{aligned} \text{After 2nd dose } A &= 10 + \frac{1}{3} \times 10 \text{ mL} \\ &= 13.33 \text{ mL} \quad (2 \text{ d.p.}) \end{aligned}$$

ii. After the 3rd dose

$$\begin{aligned} A_3 &= 10 + \frac{1}{3} \left(10 + \frac{1}{3} \times 10 \right) \\ &= 10 + \frac{1}{3} \times 10 + \left(\frac{1}{3} \right)^2 \times 10 \end{aligned}$$

$$\Rightarrow \text{GP where } a = 10, \quad r = \frac{1}{3}$$

Since $|r| < 1$:

$$\begin{aligned} S_\infty &= \frac{a}{1 - r} \\ &= \frac{10}{1 - \frac{1}{3}} \\ &= \frac{10}{\frac{2}{3}} \\ &= 15 \end{aligned}$$

$\therefore$ The amount of the drug will never exceed 15 mL.

28. Financial Maths, 2ADV M1 SM-Bank 8

Initial length = 14.2 cm

$$G_1 = 3.6$$

$$G_2 = (0.75)G_1 = (0.75)3.6$$

$$G_3 = (0.75)G_2 = (0.75^2)3.6$$

Growth is a geometric sequence

$$\underbrace{3.6, 3.6(0.75), 3.6(0.75)^2, \dots}_{\text{GP where } a=3.6, r=0.75}$$

Since $|r| < 1$,

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{3.6}{1-0.75} \\ &= 14.4 \end{aligned}$$

$\therefore$ Maximum length of fish

$$\begin{aligned} &= 14.2 + 14.4 \\ &= 28.6 \text{ cm} \end{aligned}$$

29. Financial Maths, 2ADV M1 2004 HSC 7a

$$\begin{aligned} \sum_{n=2}^4 n^2 &= 2^2 + 3^2 + 4^2 \\ &= 4 + 9 + 16 \\ &= 29 \end{aligned}$$

30. Financial Maths, 2ADV M1 SM-Bank 11

$$\begin{aligned} 0.3232\dots &= \frac{32}{100} + \frac{32}{10^4} + \frac{32}{10^6} + \dots \\ &= \frac{32}{100} \left(1 + \frac{1}{10^2} + \frac{1}{10^4} + \dots \right) \\ &\Rightarrow \text{GP where } a = 1, r = \frac{1}{10^2} \\ &= \frac{32}{100} \left(\frac{a}{1-r} \right) \\ &= \frac{32}{100} \left(\frac{1}{1 - \frac{1}{100}} \right) \\ &= \frac{32}{100} \left(\frac{1}{\frac{99}{100}} \right) \\ &= \frac{32}{99} \end{aligned}$$

31. Financial Maths, 2ADV M1 2018 HSC 14d

i. Day 1: $2^1 + 1 = 3$

Day 2: $2^2 + 2 = 6$

Day 3: $2^3 + 3 = 11$

ii. Total downloads (20 days)

$$\begin{aligned} &= 2^1 + 1 + 2^2 + 2 + \dots + 2^{20} + 20 \\ &= \underbrace{2^1 + 2^2 + \dots + 2^{20}}_{\text{GP, } a=2, r=2} + \underbrace{1 + 2 + \dots + 20}_{\text{AP, } a=1, d=1} \\ &= \frac{2(2^{20} - 1)}{2 - 1} + \frac{20}{2}(1 + 20) \\ &= 2\,097\,150 + 210 \\ &= 2\,097\,360 \end{aligned}$$

♦ Mean mark 38%.

32. Financial Maths, 2ADV M1 2008 HSC 5b

i. Limiting sum when $|r| < 1$

$$r = \frac{T_2}{T_1} = \frac{10x}{5} = 2x$$

$$\therefore |2x| < 1$$

$$\text{If } 2x > 0 \quad \text{If } 2x < 0$$

$$2x < 1 \quad -(2x) < 1$$

$$x < \frac{1}{2} \quad 2x > -1$$

$$x > -\frac{1}{2}$$

$$\therefore \text{Limiting sum when } -\frac{1}{2} < x < \frac{1}{2}$$

ii. Given $S_\infty = 100$, find x

$$\Rightarrow S_\infty = \frac{a}{1-r} = 100$$

$$\frac{5}{1-2x} = 100$$

$$100(1-2x) = 5$$

$$200x = 95$$

$$\therefore x = \frac{95}{200} = \frac{19}{40}$$

33. Financial Maths, 2ADV M1 2011 HSC 5a

i. $T_1 = a = 27$

$$T_2 = 27 \times 2^1 = 54$$

$$T_3 = 27 \times 2^2 = 108$$

$$\Rightarrow \text{GP where } a = 27, r = 2$$

$$\vdots$$

$$T_n = ar^{n-1}$$

$$T_{12} = 27 \times 2^{11} = 55\,296$$

$\therefore$ On Day 12, there are 55 296 members.

ii. Find n such that $T_n > 10\,000\,000$

$$T_n = 27(2^{n-1})$$

$$27 \times 2^{n-1} > 10\,000\,000$$

$$2^{n-1} > \frac{10\,000\,000}{27}$$

$$\ln 2^{n-1} > \ln \left(\frac{10\,000\,000}{27} \right)$$

$$(n-1)\ln 2 > \ln(370\,370.370)$$

$$n-1 > \frac{\ln(370\,370.370)}{\ln 2}$$

$$n-1 > 18.499\dots$$

$$n > 19.499\dots$$

$\therefore$ On the 20th day, the number of members $> 10\,000\,000$.

iii. If the site earns 0.5 cents per day per member,

$$\text{On Day 1, it earns } 27 \times 0.5 = 13.5 \text{ cents}$$

$$\text{On Day 2, it earns } 27 \times 2 \times 0.5 = 27 \text{ cents}$$

$$T_1 = a = 13.5$$

$$T_2 = 27$$

$$T_3 = 54$$

$$\Rightarrow \text{GP where } a = 13.5, r = 2$$

S_{12} = the total amount of money earned in the first 12 Days

MARKER'S COMMENT: Better responses stated the general term, T_n before any substitution was made, as shown in the worked solutions.

MARKER'S COMMENT: Many elementary errors were made by students in dealing with logarithms. *BE VIGILANT.*

$$\begin{aligned}
 S_{12} &= \frac{a(r^n - 1)}{r - 1} \\
 &= \frac{13.5(2^{12} - 1)}{2 - 1} \\
 &= 55\,282.5 \text{ cents} \\
 &= 552.825 \text{ dollars}
 \end{aligned}$$

♦ Mean mark 44%.
NOTE: This question can also be easily solved by making S_{12} the total sum of members (each day) and then multiplying by 0.5 cents.

∴ The site earned \$553 in the first 12 Days (nearest \$).

34. Financial Maths, 2ADV M1 2013 HSC 12c

i. Let K_n = Kim's salary in Year n

$$\left. \begin{aligned} K_1 &= a = 30\,000 \\ K_2 &= ar^1 = 30\,000(1.05^1) \end{aligned} \right\} \Rightarrow GP \quad \begin{aligned} a &= 30\,000 \\ r &= 1.05 \end{aligned}$$

⋮

$$\therefore K_{10} = ar^9 = 30\,000(1.05)^9 = \$46\,539.85$$

Let A_n = Alex's salary in Year n

$$\left. \begin{aligned} A_1 &= a = 33\,000 \\ A_2 &= 33\,000 + 1500 = 34\,500 \end{aligned} \right\} \Rightarrow AP \quad \begin{aligned} a &= 33\,000 \\ d &= 1500 \end{aligned}$$

⋮

$$A_{10} = a + 9d = 33\,000 + 1500(9) = \$46\,500$$

$$\Rightarrow K_{10} > A_{10}$$

∴ Kim earns more than Alex in the 10th year

ii. In the first 10 years, Kim earns

$$K_1 + K_2 + \dots + K_{10}$$

$$\begin{aligned}
 S_{10} &= a \left(\frac{r^n - 1}{r - 1} \right) \\
 &= 30\,000 \left(\frac{1.05^{10} - 1}{1.05 - 1} \right) \\
 &= 377\,336.78
 \end{aligned}$$

∴ In the first 10 years, Kim earns \$377 336.78

iii. Let T_n = Alex's savings in Year n

$$\left. \begin{aligned} T_1 &= a = \frac{1}{3}(33\,000) = 11\,000 \\ T_2 &= a + d = \frac{1}{3}(34\,500) = 11\,500 \\ T_3 &= a + 2d = \frac{1}{3}(36\,000) = 12\,000 \end{aligned} \right\} \Rightarrow AP \quad \begin{aligned} a &= 11\,000 \\ d &= 500 \end{aligned}$$

Find n such that $S_n = 87\,500$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$87\,500 = \frac{n}{2}[22\,000 + (n-1)500]$$

$$87\,500 = \frac{n}{2}[21\,500 + 500n]$$

$$250n^2 + 10\,750n - 87\,500 = 0$$

$$n^2 + 43n - 350 = 0$$

$$(n-7)(n+50) = 0$$

$$\therefore n = 7, \quad n > 0$$

$\therefore$ Alex's savings will be \$87,500 after 7 years.

♦ Mean mark 45%.

IMPORTANT: Using the AP sum formula to create and then solve a quadratic in n is challenging and often examined. Students need to solve and *interpret* the solutions.

35. Trigonometry, 2ADV T2 2004 HSC 9a

i. $1 - \tan^2 \theta + \tan^4 \theta - \dots$

$$\Rightarrow \text{GP where } a = 1, \quad r = \frac{T_2}{T_1} = -\tan^2 \theta$$

$$\therefore S_\infty = \frac{1}{1 - (-\tan^2 \theta)}$$

$$= \frac{1}{1 + \tan^2 \theta}$$

$$= \frac{1}{\sec^2 \theta}$$

$$= \cos^2 \theta$$

ii. Find θ such that $|r| < 1$

$$|-\tan^2 \theta| < 1$$

$$\tan^2 \theta < 1$$

$$-1 < \tan \theta < 1$$

$$\therefore -\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

36. Financial Maths, 2ADV M1 2016 HSC 14d

$$1 + x + x^2 + x^3 + x^4$$

$$\Rightarrow \text{GP where } a = 1, \quad r = x, \quad n = 5$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_5 = \frac{(x^5 - 1)}{x - 1}$$

$$\therefore \lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1}$$

$$= \lim_{x \rightarrow 1} (1 + x + x^2 + x^3 + x^4)$$

$$= 5$$

♦ Mean mark 45%.

37. Financial Maths, 2ADV M1 2017 HSC 16b

$$S_\infty = \frac{a}{1 - r}$$

$$\text{If } S_\infty = 2:$$

$$2 = \frac{a}{1 - r}$$

$$2(1 - r) = a$$

$$1 - r = \frac{a}{2}$$

$$r = 1 - \frac{a}{2}$$

$$\text{Since } |r| < 1,$$

$$\left|1 - \frac{a}{2}\right| < 1$$

$$1 - \frac{a}{2} < 1 \quad \text{or} \quad -\left(1 - \frac{a}{2}\right) < 1$$

$$\frac{a}{2} > 0$$

$$a > 0$$

$$\frac{a}{2} < 2$$

$$a < 4$$

$$\therefore 0 < a < 4$$

♦ Mean mark 39%.

38. Financial Maths, 2ADV M1 2011 HSC 9d

$$\begin{aligned} \text{i. } & \frac{1}{\sqrt{n} + \sqrt{n+1}} \times \frac{\sqrt{n} - \sqrt{n+1}}{\sqrt{n} - \sqrt{n+1}} \\ &= \frac{\sqrt{n} - \sqrt{n+1}}{(\sqrt{n})^2 - (\sqrt{n+1})^2} \\ &= \frac{\sqrt{n} - \sqrt{n+1}}{n - (n+1)} \\ &= \frac{\sqrt{n} - \sqrt{n+1}}{-1} \\ &= \sqrt{n+1} - \sqrt{n} \end{aligned}$$

MARKER'S COMMENT: Students connecting the information between parts (i) and (ii) were easily the most successful. **AGAIN**, the information from earlier parts of multi-part questions is **gold plated information** for directing your answer.

$$\begin{aligned} \text{ii. } & \frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{99} + \sqrt{100}} \\ &= (\sqrt{2} - \sqrt{1}) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \dots + (\sqrt{100} - \sqrt{99}) \\ &= -\sqrt{1} + \sqrt{100} \\ &= -1 + 10 \\ &= 9 \end{aligned}$$

♦♦♦ Mean mark 15%.

39. Probability, 2ADV S1 2013 HSC 15d

i. $P(\text{Pat wins on 1st throw}) = P(W)$

$$\begin{aligned} P(W) &= P(\text{Pat throws 2 sixes}) \\ &= \frac{1}{6} \times \frac{1}{6} \\ &= \frac{1}{36} \end{aligned}$$

ii. Let $P(L) = P(\text{loss for either player on a throw}) = \frac{35}{36}$

$$\begin{aligned} & P(\text{Pat wins on 1st or 2nd throw}) \\ &= P(W) + P(LLW) \\ &= \frac{1}{36} + \left(\frac{35}{36}\right) \times \left(\frac{35}{36}\right) \times \left(\frac{1}{36}\right) \\ &= \frac{2521}{46656} \\ &= 0.054 \text{ (to 3 d.p.)} \end{aligned}$$

♦♦ Mean mark 33%

MARKER'S COMMENT: Many students did not account for Chandra having to lose when Pat wins on the **2nd** attempt.

iii. $P(\text{Pat wins eventually})$

$$\begin{aligned} &= P(W) + P(LLW) + P(LLLLW) + \dots \\ &= \frac{1}{36} + \left(\frac{35}{36}\right)^2 \left(\frac{1}{36}\right) + \left(\frac{35}{36}\right)^4 \left(\frac{1}{36}\right) + \dots \\ \Rightarrow \text{GP where } a &= \frac{1}{36}, r = \left(\frac{35}{36}\right)^2 = \frac{1225}{1296} \end{aligned}$$

Since $|r| < 1$:

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{\frac{1}{36}}{1 - \left(\frac{1225}{1296}\right)} \\ &= \frac{1}{36} \times \frac{1296}{71} \\ &= \frac{36}{71} \end{aligned}$$

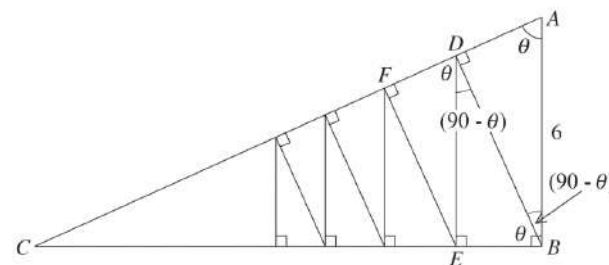
♦♦♦ Mean mark 8%!

COMMENT: Be aware that diminishing probabilities and S_{∞} within the Series and Applications are a natural cross-topic combination.

$\therefore$ Pat's chances to win eventually are $\frac{36}{71}$.

40. Trigonometry, 2ADV T1 2005 HSC 9b

i.



Show $EF = 6 \sin^3 \theta$

In $\triangle ADB$

$$\sin \theta = \frac{DB}{6}$$

$$DB = 6 \sin \theta$$

$$\angle ABD = 90 - \theta \quad (\text{angle sum of } \triangle ADB)$$

$$\therefore \angle DBE = \theta \quad (\angle ABE \text{ is a right angle})$$

In $\triangle BDE$:

$$\sin \theta = \frac{DE}{DB}$$

$$= \frac{DE}{6 \sin \theta}$$

$$DE = 6 \sin^2 \theta$$

$$\angle BDE = 90 - \theta \quad (\text{angle sum of } \triangle DBE)$$

$$\angle EDF = \theta \quad (\angle FDB \text{ is a right angle})$$

In $\triangle DEF$:

$$\sin \theta = \frac{EF}{DE}$$

$$= \frac{EF}{6 \sin^2 \theta}$$

$$\therefore EF = 6 \sin^3 \theta \quad \dots \text{as required}$$

ii. Show $BD + EF + GH \dots$

$$\text{has limiting sum} = 6 \sec \theta \tan \theta$$

$$\underbrace{6 \sin \theta + 6 \sin^3 \theta + \dots}_{\text{GP where } a=6 \sin \theta, \ r=\sin^2 \theta}$$

Since $0 < \theta < 90^\circ$

$$-1 < \sin \theta < 1$$

$$0 < \sin^2 \theta < 1$$

$$\therefore |r| < 1$$

$$\therefore S_\infty = \frac{a}{1-r}$$

$$= \frac{6 \sin \theta}{1 - \sin^2 \theta}$$

$$= \frac{6 \sin \theta}{\cos^2 \theta}$$

$$= 6 \times \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta}$$

$$= 6 \sec \theta \tan \theta \dots \text{as required.}$$